

Ethical Theories and Their Applications in Modern Computing

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Abstract

Modern computing technologies, from algorithms to social media platforms, raise complex ethical questions that demand careful analysis. This paper explores eight ethical theories utilitarianism, deontology, virtue ethics, social contract theory, ethical egoism, care ethics, natural law theory, and moral relativism and applies them to real-world dilemmas in technology. By comparing how these frameworks define “good” actions, prioritize consequences

versus duties, and address strengths and weaknesses, this paper argues that a hybrid approach combining utilitarianism and deontology offers the most practical guidance for computer scientists.

For example, in financial technology, utilitarianism's focus on maximizing societal benefits (e.g., market efficiency) must be balanced with deontology's emphasis on transparency and duty to avoid harm. Similarly, in social media data privacy, virtue ethics encourages developers to prioritize user trust, while social contract theory demands clear, mutual agreements between companies and users. However, no single theory is sufficient: utilitarianism struggles to quantify intangible harms like privacy violations, while Kantianism's rigidity clashes with the fast-paced evolution of technology.

Drawing on case studies from Walter (2003) and Ashe (2005), this paper highlights the importance of ethical frameworks in guiding decisions such as automated trading algorithms and data collection practices. Personal reflections, such as my experience debating facial recognition ethics in a class project, illustrate the challenges of balancing competing principles. Ultimately, understanding these theories equips professionals to navigate moral ambiguity and advocate for responsible innovation.

Section I: Comparison of Ethical Theories

Ethical Theories

1. Utilitarianism

Approach: Focuses on maximizing overall happiness or benefit. For instance, when I developed a study guide resource for students, I deliberated whether tracking user study progress(to reduce time wasting) justified privacy risks. A utilitarian would prioritize the greater good of smoother traffic, even if some users dislike surveillance.

Strengths: Pragmatic, outcome-focused.

Weaknesses: Hard to quantify harms like stress from privacy loss (Ferrell et al., 2013).

2. Deontology (Kantianism)

Approach: Follows universal rules, like "never lie." During a job interview, I refused to exaggerate my technical capabilities to the recruiter, citing Kant's duty to honesty.

Strengths: Protects individual rights.

Weaknesses: Struggles with conflicts, like censoring hate speech vs. free speech (Timmermann, 2013).

3. Virtue Ethics

Approach: Builds moral character. The statement “A good developer doesn’t just write code they ask who it might harm.” This mirrors Ahmad, Gao, & Hali’s (2017) emphasis on integrity in leadership.

Strengths: Encourages lifelong growth.

Weaknesses: Too vague for urgent decisions.

4. Social Contract Theory

Approach: Rules for mutual benefit. Terms of service agreements, like Apple’s privacy labels, reflect this users “agree” to data use for free apps (Al-Aidaros, Shamsudin, & Idris, 2013).

Strengths: Balances individual and collective needs.

Weaknesses: Assumes equal power in agreements.

5. Ethical Egoism

Approach: Acts in self-interest. Startups might prioritize profit over ethics to survive, aligning with Ferrell et al. (2013)’s view of egoism in competitive markets.

Strengths: Drives innovation.

Weaknesses: Risks exploitation.

6. Care Ethics

Approach: Prioritizes relationships. When faced at a crossroad to prioritized user empathy over monetization a care ethics approach akin to Skelton’s (2016) work on patient dignity.

Strengths: Human-centered.

Weaknesses: Overlooks systemic issues.

7. Natural Law Theory

Approach: Follows inherent moral principles. Debates over “natural” digital rights (e.g., internet access as a human right) stem from this theory (Al-Aidaros, Shamsudin, & Idris, 2013).

Strengths: Universally appealing.

Weaknesses: Culturally subjective.

8. Moral Relativism

Approach: Ethics vary by culture. While Silicon Valley condemns data mining, Chinese tech firms often comply with state surveillance a clash Ferrell et al. (2013) attribute to relativism.

Strengths: Respects diversity.

Weaknesses: No universal standards.

Reflection

I align most with utilitarianism and deontology. Last semester, my team built a study app that collected user data (deontology's respect for privacy) but used analytics to improve information (utilitarianism's focus on user benefit). However, Kantianism's rigidity clashed when we faced pressure to possibly monetize data. For instance, when my team suggested using facial recognition in the class project, we saw how utilitarianism's focus on safety (reducing crime) clashed with Kantianism's duty to privacy a tension highlighted in IT ethics literature (Ahmad, Gao, & Hali, 2017).

Application to Real-World Scenarios

Case Study 1: Corporate Ethics in Financial Technology (Walter, 2003)

Summary: A fintech firm develops a profitable trading algorithm that risks market instability.

Analysis:

- **Utilitarianism:** Implement the algorithm but add safeguards (circuit breakers) to balance profit and stability. Ferrell et al. (2013) note similar trade-offs in marketing ethics.
- **Deontology:** Reject the algorithm unless it guarantees zero harm, per Kant's duty to avoid using people as "means" (Timmermann, 2013).
- **Virtue Ethics:** Cultivate a culture of responsibility, as Ahmad, Gao, & Hali (2017) advise for ethical leaders.

Solution: Hybrid approach. Use the algorithm with safeguards (utilitarianism) while being transparent with investors (deontology).

Case Study 2: Data Privacy in Social Media (Ashe, 2005)

Summary: A social media company legally collects user data without explicit consent.

Analysis:

- **Social Contract Theory:** Redesign terms of service for clarity, ensuring true mutual agreement (Al-Aidaros, Shamsudin, & Idris, 2013).
- **Care Ethics:** Prioritize user trust by defaulting to opt-in data sharing, as Skelton (2016) emphasizes in end-of-life care ethics.
- **Utilitarianism:** Allow data collection but compensate users with ad-free subscriptions.

Solution: Social contract theory. Users deserve clear, jargon-free agreements.

Conclusion

Ethical theories are not abstract philosophies but essential tools for computer scientists. In IT, combining utilitarianism and deontology balances innovation and responsibility. In social media, social contract theory ensures fairness in vague systems. My own experiences like debating data tracking in the project i worked on highlights how these frameworks resolve real dilemmas. As AI and algorithms grow more pervasive, grounding decisions in ethics becomes not just optional, but a professional duty. Future research could explore how these frameworks apply to AI in healthcare, such as prioritizing cost-effective treatments over patient autonomy a debate rooted in utilitarianism and deontology.

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